

Lecture 8: Planetary Interiors Structure, Composition, & Dynamics

Planetary Systems

Tim Lichtenberg

Kapteyn Astronomical Institute
University of Groningen

2026



Learning objectives

By the end of this lecture, you will be able to:

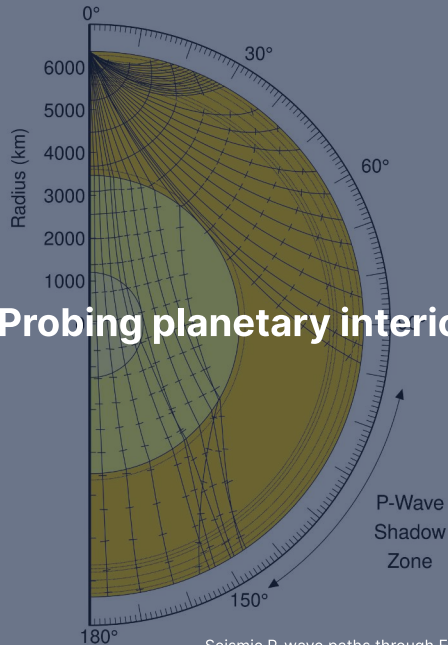
- Explain how seismology, gravity, and moment of inertia probe interiors
- Derive C/MR^2 for uniform and two-layer planets
- Apply hydrostatic equilibrium and equations of state
- Compare internal structures across the solar system
- Describe recent interior results from InSight seismology

Recap: Lecture 7

- Crater scaling laws and surface chronology from crater counting
- Volcanism, tectonics, and erosion styles across the solar system
- Remote sensing and cryovolcanism on outer icy worlds

Today: What lies *beneath* the surface? How do we know, and what does it tell us?

1. Probing planetary interiors



Seismic P-wave paths through Earth. Credit: Wikimedia Commons, public domain

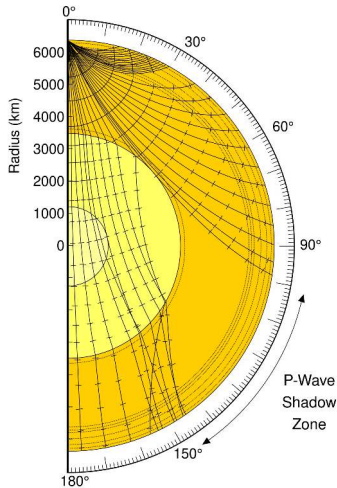
Three ways to see inside a planet

We cannot directly sample deep interiors. Three complementary tools:

1. **Seismology:** wave speeds trace density and elastic moduli
2. **Gravity fields:** spacecraft tracking measures mass distribution (J_2)
3. **Moment of inertia:** degree of central mass concentration (C/MR^2)

For Earth, all three apply; for other worlds, gravity and C/MR^2 are often our only probes.

Seismic waves: P-waves and S-waves



P-waves (primary, compressional):

- Travel through solids *and* liquids
- $v_p = \sqrt{(K + \frac{4}{3}\mu)/\rho}$

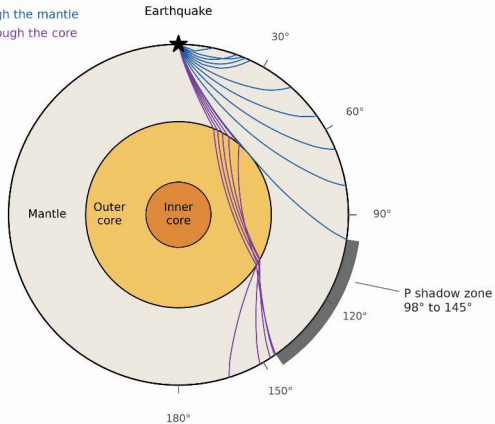
S-waves (secondary, shear):

- Travel through solids *only*
- $v_s = \sqrt{\mu/\rho}$
- $\mu = 0$ in a liquid $\Rightarrow$ S-waves **cannot propagate**

Credit: Wikimedia Commons, public domain

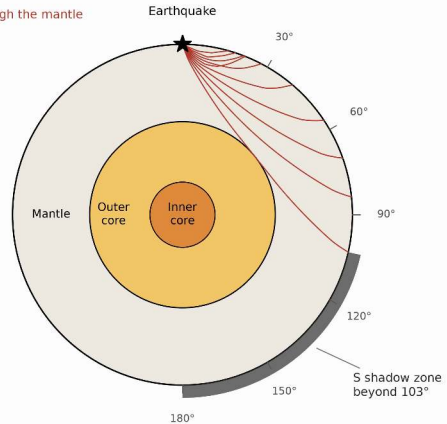
(a) P waves: refracted into the core

P through the mantle
PKP through the core



(b) S waves: stopped at the liquid core

S through the mantle



Rays traced through PREM, the standard seismological model of Earth's interior. The absence of S-waves beyond 103° proves the outer core is liquid, while P-wave refraction creates a shadow zone from 103° to 143°. Course original, from the PREM tabulation of Dziewonski & Anderson (1981).

Gravity field measurements

Planetary rotation produces an oblate shape whose gravitational quadrupole J_2 is measured via spacecraft Doppler tracking.

- Rotational flattening produces a quadrupole potential term J_2
- Spacecraft radio tracking detects minute Doppler velocity shifts
- Juno and GRAIL constrain internal mass distribution to high precision

The moment of inertia factor: C/MR^2

The moment of inertia factor measures central mass concentration:

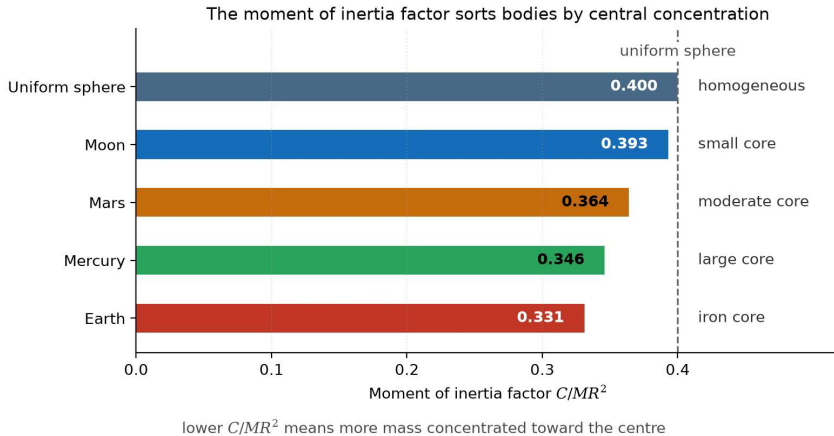
Darwin-Radau relation:

$$\frac{C}{MR^2} = \frac{2}{3} \left[1 - \frac{2}{5} \sqrt{\frac{4 - k_f}{1 + k_f}} \right]$$

- Hydrostatic bodies: from the flattening, $k_f = 3J_2/m$ with $m = \omega^2 R^3 / GM$
- Moon, Mars, Mercury: from J_2 and the precession or libration rate
- Lower value $\Rightarrow$ denser core

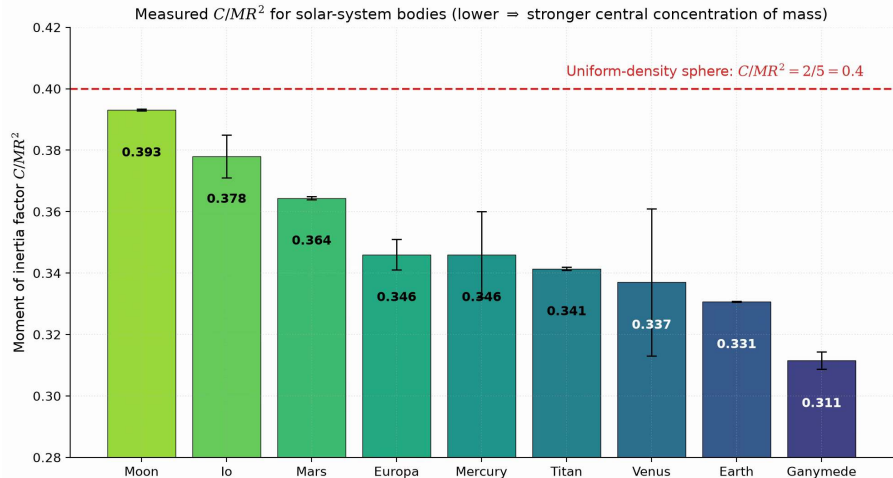
Body	C/MR^2	State
Uniform sphere	0.400	Homogeneous
Moon	0.393	Small core
Mars	0.364	Moderate core
Mercury	0.346	Large core
Earth	0.331	Iron core

Source: de Pater & Lissauer (2010)



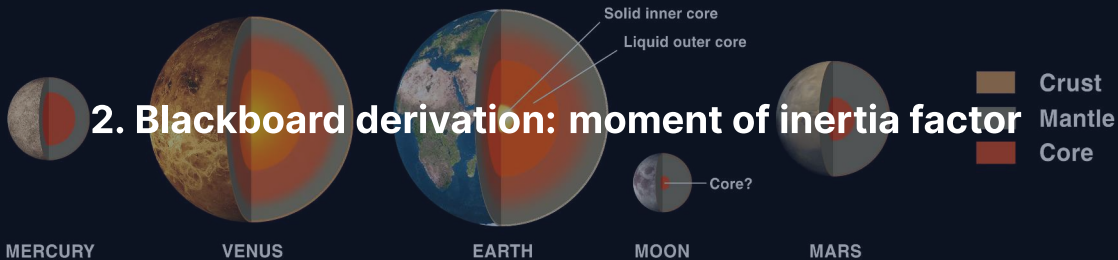
The moment of inertia factor C/MR^2 : a uniform sphere gives 0.400; the Moon (0.393), Mars (0.364), Mercury (0.346) and Earth (0.331) fall below it in the order of their central mass concentration, from a small core to a large iron core. Values after de Pater & Lissauer (2010).

Course figure.

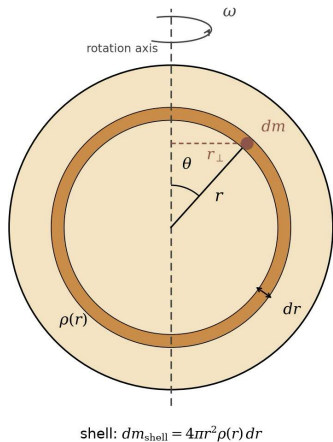


Moment of inertia factors for nine solar-system bodies. Every body falls below the uniform-sphere value of 0.400 because central mass concentration lowers C/MR^2 . Course original; lunar value from Williams et al. (2014).

2. Blackboard derivation: moment of inertia factor



Deriving C/MR^2



Goal: Derive the moment of inertia factor for (a) a uniform sphere and (b) a two-layer differentiated body.

Starting point: For a spherically symmetric body, the moment of inertia about the rotation axis is:

$$C = \frac{8\pi}{3} \int_0^R \rho(r) r^4 dr$$

A mass element at colatitude θ contributes $r_{\perp}^2 dm = r^2 \sin^2 \theta dm$; averaged over a thin shell, $\sin^2 \theta \rightarrow 2/3$.

Credit: Course original

Summary of derived results

Uniform sphere: $\rho = \text{const}$

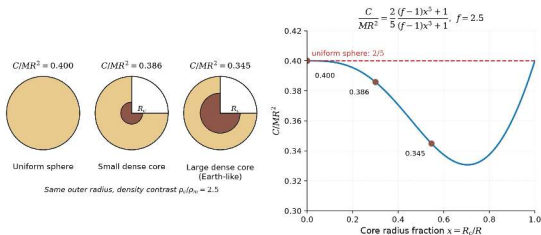
$$C = \frac{8\pi}{15}\rho R^5, \quad M = \frac{4}{3}\pi\rho R^3 \quad \Rightarrow \quad \frac{C}{MR^2} = \frac{2}{5} = 0.400$$

Two-layer body: core (ρ_c, R_c) and mantle (ρ_m, R), with $x \equiv R_c/R$ and $f \equiv \rho_c/\rho_m$:

$$\frac{C}{MR^2} = \frac{2}{5} \cdot \frac{(f-1)x^5 + 1}{(f-1)x^3 + 1}$$

Since $x^5 < x^3$ for $0 < x < 1$, any dense core ($f > 1$) pulls $C/MR^2 < 0.400$.

Why differentiation lowers C/MR^2

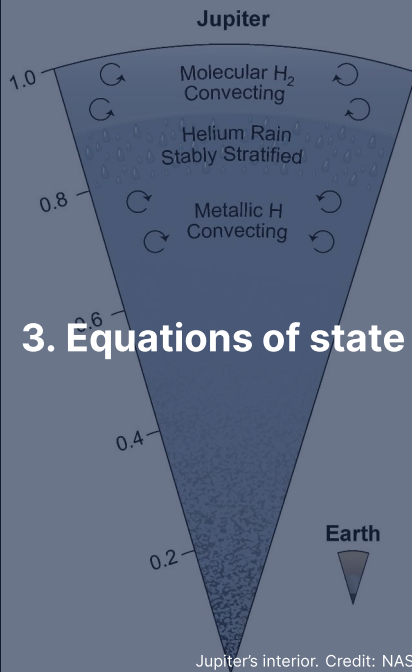


Credit: Course original

C weights mass by r^2 : mass moved inward lowers C but not M .

- Fixed $f = \rho_c/\rho_m = 2.5$; core radius fraction $x = R_c/R$ varies
- Earth ($x = 0.546$) gives $C/MR^2 \approx 0.345$; measured 0.331, the gap is self-compression

Any dense core pulls C/MR^2 below 0.400.



Hydrostatic equilibrium in spherical symmetry

Pressure at each depth supports the weight of overlying material:

$$\frac{dP}{dr} = -\rho(r) g(r) = -\frac{G m(r) \rho(r)}{r^2}$$

where $m(r) = 4\pi \int_0^r \rho(r') r'^2 dr'$ is the mass enclosed within radius r .

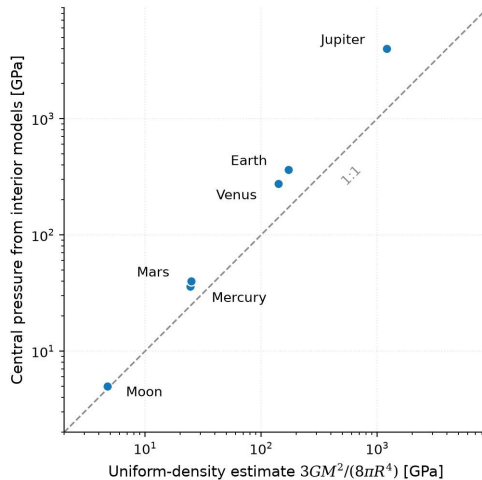
Combined with an equation of state $P = P(\rho, T)$, this integrates inward to give the pressure-density profile.

Central pressure estimate

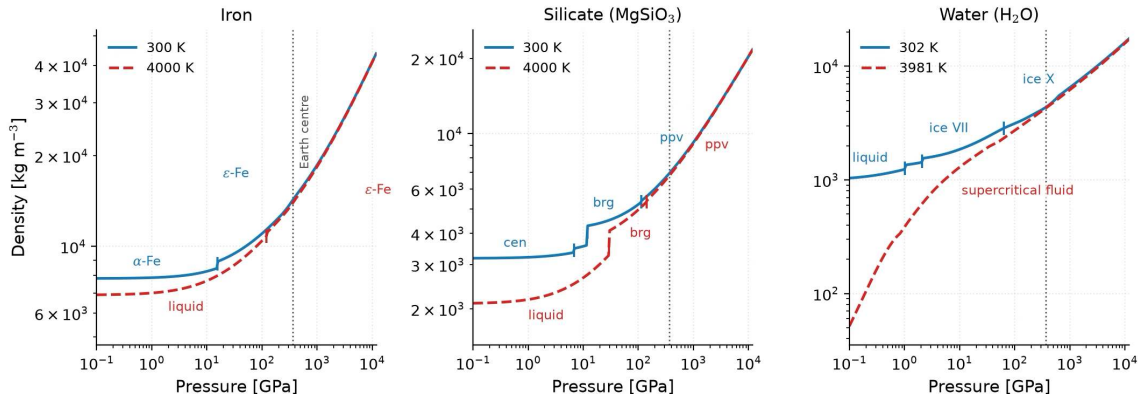
Uniform density gives an analytical central pressure; it is a lower bound for a differentiated body.

$$P_c \approx \frac{3 G M^2}{8\pi R^4}$$

- Earth: estimate ~170 GPa vs. actual ~360 GPa
- Mars: estimate ~25 GPa vs. actual ~40 GPa
- Uniform estimate is always too low because real planets are centrally concentrated



Central pressure from interior models against the uniform-density estimate for six solar-system bodies. Real planets sit above the equality line because density increases inward, from a few per cent for the Moon to 3× for Jupiter. Course original.



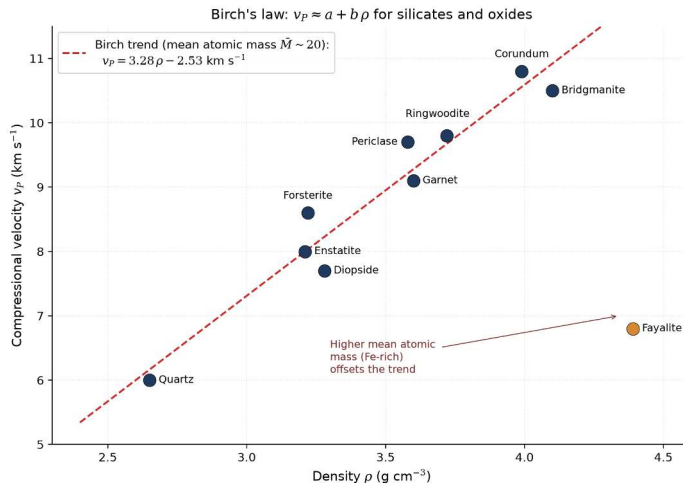
Multiphase equations of state for iron, silicate, and water: density against pressure for a cold (~300 K) and a hot (~4000 K) isotherm. Condensed matter is stiff under planetary pressures, with discrete density jumps marking phase transitions. Course original, from the PAEOS tables of Attia et al. (2026).

Birch's law

Birch's law provides an empirical linear bridge from measured seismic wave speeds directly to interior density.

$$v_p = a(\bar{M}) + b \rho$$

- Linear relation between compressional velocity v_p and density ρ
- Parameters depend primarily on mean atomic weight $\bar{M}$
- Breaks down across compositional boundaries and phase transitions



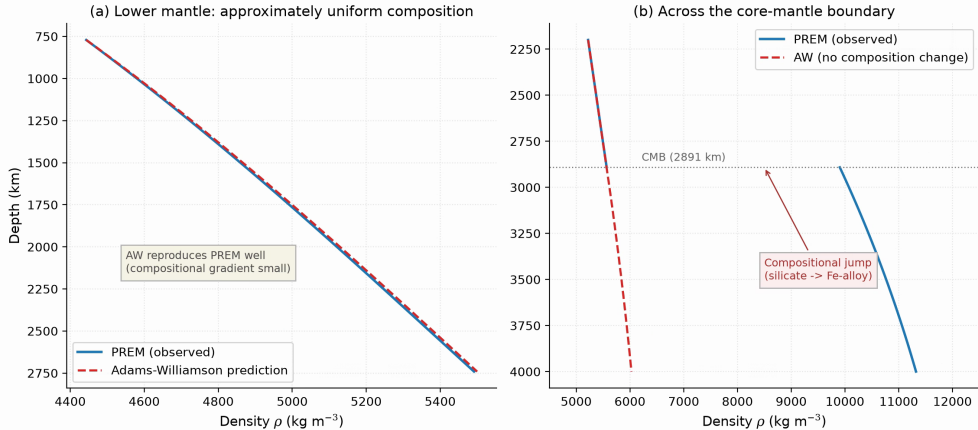
Birch's law: v_p against density for mantle minerals. Compressional velocity scales linearly with density at constant mean atomic mass, while iron-rich minerals plot below the trend. Course original; trend after Birch (1952).

Adams-Williamson equation

The Adams-Williamson equation integrates seismic wave speeds to determine density profiles under adiabatic self-compression.

$$\frac{d\rho}{dr} = -\frac{\rho(r)g(r)}{\phi(r)}, \quad \phi = v_P^2 - \frac{4}{3}v_S^2$$

- Seismic parameter $\phi \equiv K_S/\rho = v_P^2 - \frac{4}{3}v_S^2$
- Assumes hydrostatic equilibrium, adiabatic gradient, and uniform composition
- Discontinuities mark phase changes or compositional boundaries (e.g., at the core-mantle boundary)

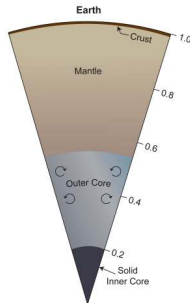


The Adams-Williamson relation tested against PREM. Adiabatic self-compression matches the lower mantle, but the sharp density jump at the core-mantle boundary exposes a change in composition. Course original, from the PREM tabulation of Dziewonski & Anderson (1981).



4. Earth's interior

Earth's internal structure



Credit: NASA/JPL-Caltech

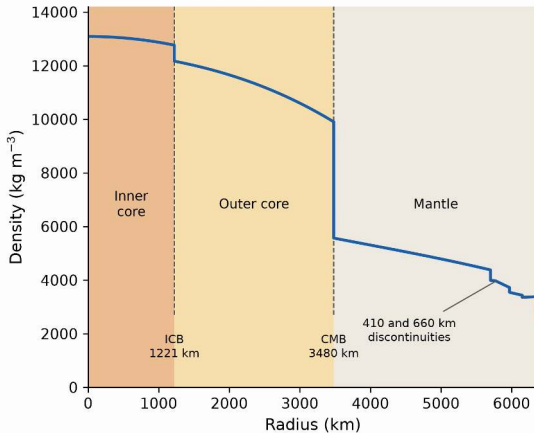
Four major layers:

1. **Crust** (0–70 km)
2. **Mantle** (Moho to 2891 km)
3. **Outer core** (2891–5150 km): liquid
4. **Inner core** (5150–6371 km): solid

Solid inner core: at 330–360 GPa, melting point exceeds 5000–6000 K.

Mean density	5515 kg m^{-3}
C/MR^2	0.3307
Central pressure	$\sim 360 \text{ GPa}$

The PREM density profile



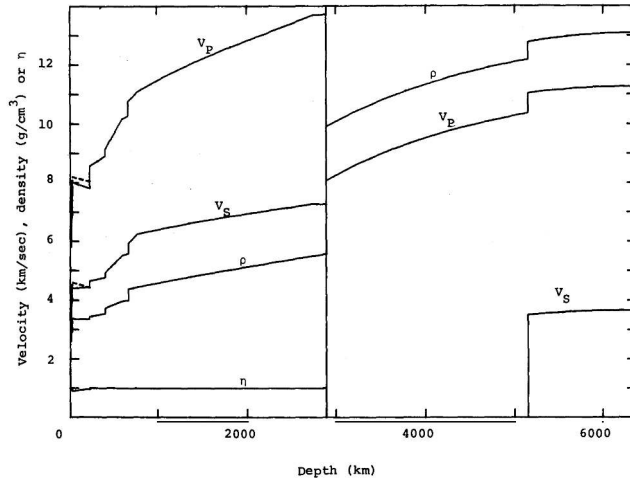
Credit: Dziewonski & Anderson (1981)

Preliminary Reference Earth Model (PREM):

- Upper mantle: $\sim 3,300 \text{ kg m}^{-3}$
- Lower mantle base: $\sim 5,500 \text{ kg m}^{-3}$
- Outer core: $\sim 9,900\text{--}12,200 \text{ kg m}^{-3}$
- Inner core: $\sim 13,000 \text{ kg m}^{-3}$

Discontinuities at 410 and 660 km: phase transitions.

Core-mantle boundary ($\sim 3,480 \text{ km}$): silicate to iron.



PREM radial profiles of wave speeds, density, and anisotropy; shear velocity v_s vanishes in the liquid outer core and resumes in the solid inner core. Credit: Dziewonski & Anderson (1981), Fig. 8.

Crust, mantle, and core

Layer	Depth (km)	ρ (kg m ⁻³)	Composition
Crust	0–70	2700–3000	Silicates (Moho base)
Upper mantle	70–410	3300–3700	Olivine, pyroxene
Transition zone	410–660	3700–4400	Wadsleyite, ringwoodite
Lower mantle	660–2891	4400–5500	Bridgmanite
Outer core	2891–5150	9900–12,200	Liquid Fe-Ni
Inner core	5150–6371	~13,000	Solid Fe-Ni

Convection in the liquid outer core powers Earth's geodynamo, fed by inner-core growth: crystallising iron releases latent heat and light elements.



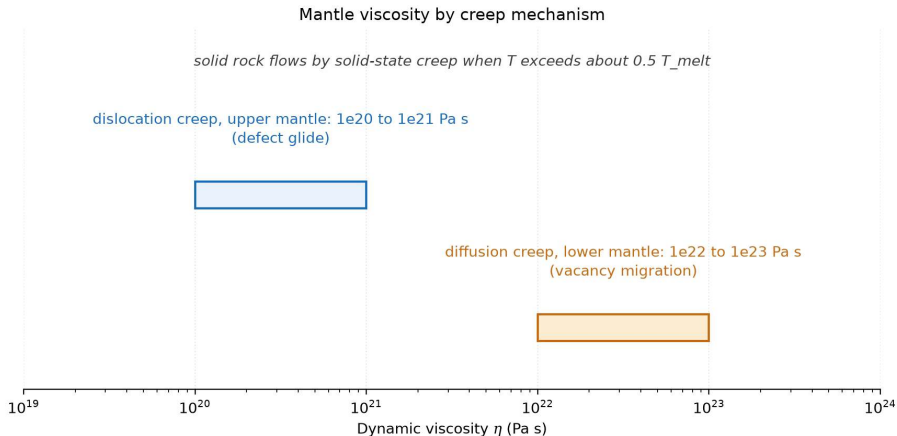
5. Mantle rheology & convection

Global surface heat flow, the surface expression of mantle convection. Credit: Fabio Crameri, Wikimedia Commons, CC BY-SA 4.0

Solid-state creep

At high homologous temperatures the crystalline silicate mantle deforms by solid-state creep, behaving as a viscous fluid over geological time.

- Solid rock flows when $T > 0.5 T_{\text{melt}}$ via atomic migration
- **Diffusion creep:** vacancy migration, dominant in lower mantle ($\eta \sim 10^{22}\text{--}10^{23}$ Pa s)
- **Dislocation creep:** defect glide, dominant in upper mantle ($\eta \sim 10^{20}\text{--}10^{21}$ Pa s)



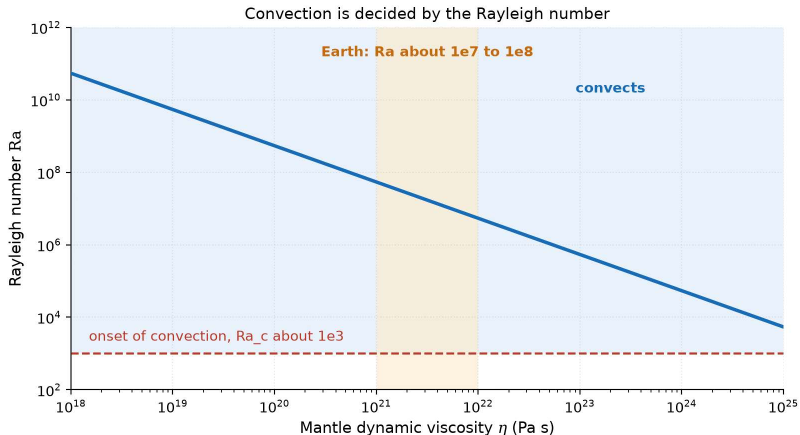
Mantle viscosity by creep mechanism: dislocation creep (defect glide) in the upper mantle at 10^{20} to 10^{21} Pa s, diffusion creep (vacancy migration) in the lower mantle at 10^{22} to 10^{23} Pa s; solid rock flows once T exceeds about $0.5 T_{\text{melt}}$. Course schematic.

The Rayleigh number

The Rayleigh number balances thermal buoyancy against viscous resistance, determining whether a planetary mantle convects.

$$Ra = \frac{\alpha \rho g \Delta T d^3}{\kappa \eta}$$

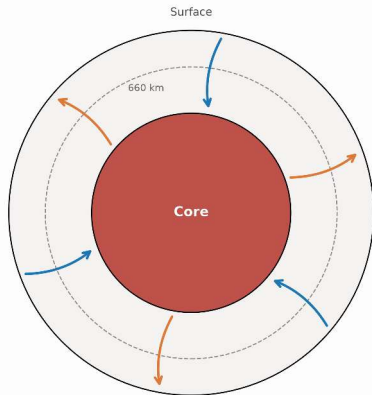
- Ra measures the ratio of buoyancy driving forces to viscous dissipation
- Critical threshold for convection onset is $Ra_c \sim 10^3$
- Earth's mantle has $Ra \sim 10^7\text{--}10^8 \gg Ra_c$, driving vigorous convection



The Rayleigh number against mantle viscosity for the course's Earth-mantle values: convection sets in above $Ra_c \approx 10^3$, and Earth's mantle at $\eta = 10^{21}$ to 10^{22} Pa s sits at $Ra \approx 10^7$ to 10^8 , four to five orders of magnitude above the threshold. Course figure.

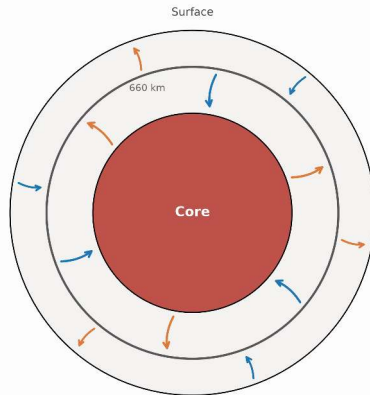
Mantle convection regimes

(a) Whole-mantle convection



Single circulation
from CMB to surface

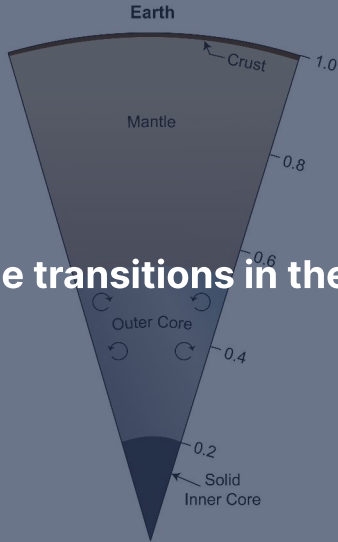
(b) Layered convection



Upper- and lower-mantle
cells separated by 660 km

End-member regimes of mantle convection. Mantle convection operates primarily in a whole-mantle regime, with the 660 km discontinuity acting as a partial, regionally variable barrier. Schematic; boundary radius exaggerated.

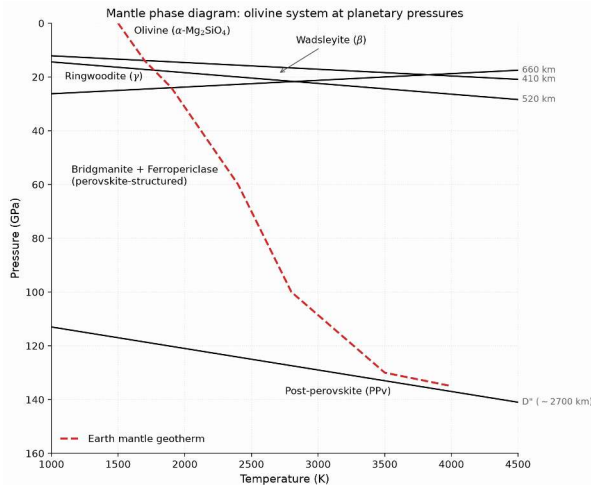
6. Phase transitions in the mantle



Mineral phase transitions with depth

Depth (km)	Transition	$\Delta\rho$
410	Olivine $\rightarrow$ wadsleyite	~5%
520	Wadsleyite $\rightarrow$ ringwoodite	~2%
660	Ringwoodite $\rightarrow$ bridgmanite + ferropericlase	~8%
~2700	Bridgmanite $\rightarrow$ post-perovskite	~1–2%

- Same chemical composition ($(\text{Mg, Fe})_2\text{SiO}_4$), denser crystal structures
- Driven by pressure: atoms rearrange into denser packing
- Transitions produce the primary seismic discontinuities in PREM



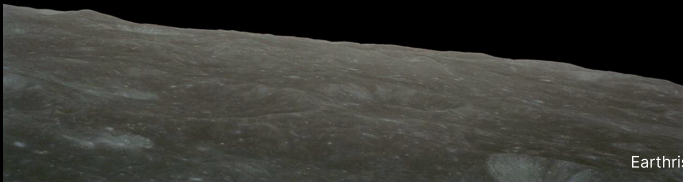
Phase boundaries of the mantle against an approximate geotherm. Seismic discontinuities occur where the geotherm crosses pressure-induced mineral phase boundaries. After Turcotte & Schubert (2002) and Murakami et al. (2004).

The 660 km discontinuity

A negative Clapeyron slope at 660 km makes the ringwoodite breakdown endothermic, creating buoyancy that resists slab penetration.

- Ringwoodite decomposes into bridgmanite plus ferropericlasite with an ~8% density jump
- Negative Clapeyron slope ($dP/dT < 0$): in a cold slab the boundary sits deeper, so buoyant ringwoodite persists below 660 km
- Produces upward buoyancy that acts as a partial barrier to whole-mantle convection

Break



Mercury's interior structure (post-MESSENGER model)

Iron core (all layers):
~83% R, ~74% mass

Crust
~30-50 km thick

Silicate mantle
~400 km thick

Solid Fe-S layer (?)
~100 km

Liquid Fe-Ni-S
outer core

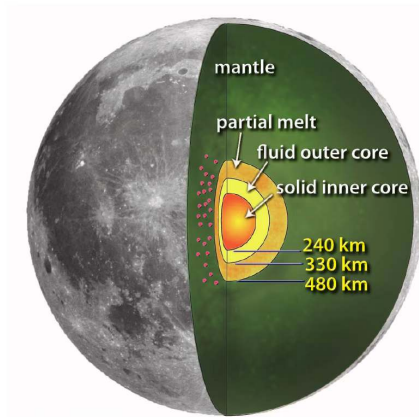
Solid inner core (?)
inferred

7. Comparative terrestrial interiors

The Moon

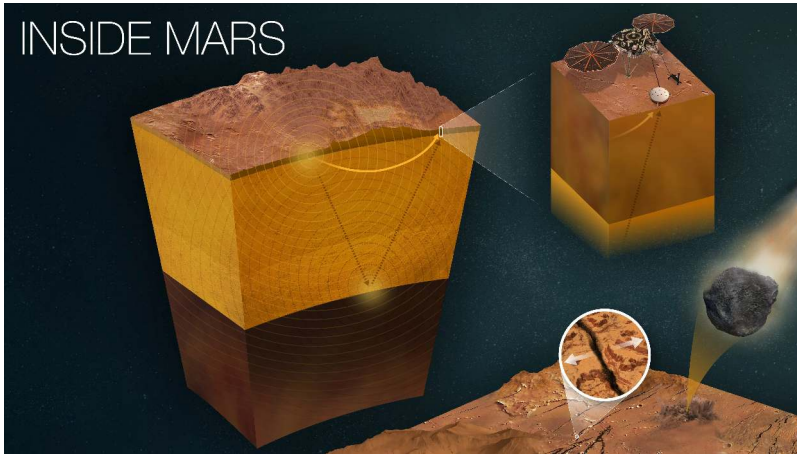
The Moon's moment of inertia factor is 0.393, close to a uniform sphere: its metallic core is exceptionally small.

- Mean density $\bar{\rho} = 3346 \text{ kg m}^{-3}$ is close to uncompressed mantle rock
- Core radius fraction is only $R_c/R \sim 0.1\text{--}0.2$, containing $< 2\%$ of lunar mass
- Low core fraction supports formation via a giant impact from Earth's mantle



The lunar interior from Apollo seismology. Reanalysed Apollo seismograms reveal a small metallic core (200–380 km radius) with a liquid outer and a solid inner part. Credit: Weber et al. (2011), Fig. 1A.

INSIDE MARS



Mars's interior as revealed by InSight seismology, showing a 24–72 km crust, a mantle to ~1560 km, and a seismic boundary at $R_c \sim 1830$ km, reinterpreted in 2023 as the top of a molten silicate layer above a smaller (~1650–1675 km) liquid iron core of about 6500 kg m^{-3} .

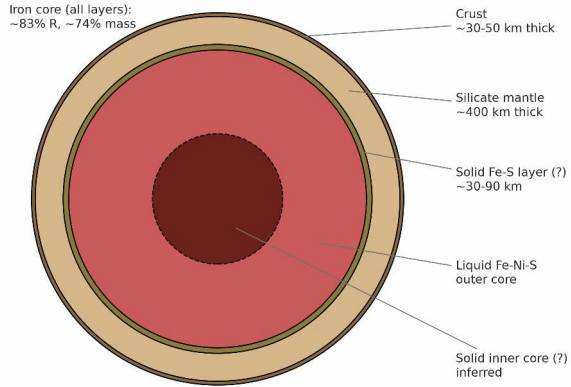
Credit: NASA/JPL-Caltech (PIA25282).

Mercury: an iron world

Mercury has the highest uncompressed density of any terrestrial planet, dominated by an enormous iron core.

- Moment of inertia factor $C/MR^2 = 0.346$ and mean density $\bar{\rho} = 5427 \text{ kg m}^{-3}$
- Metallic core reaches $R_c \sim 2000\text{--}2050 \text{ km}$ ($\sim 83\%$ of planetary radius)
- Core accounts for $\sim 74\%$ of total mass; weak dynamo indicates a liquid outer core

Mercury's interior structure (post-MESSENGER model)



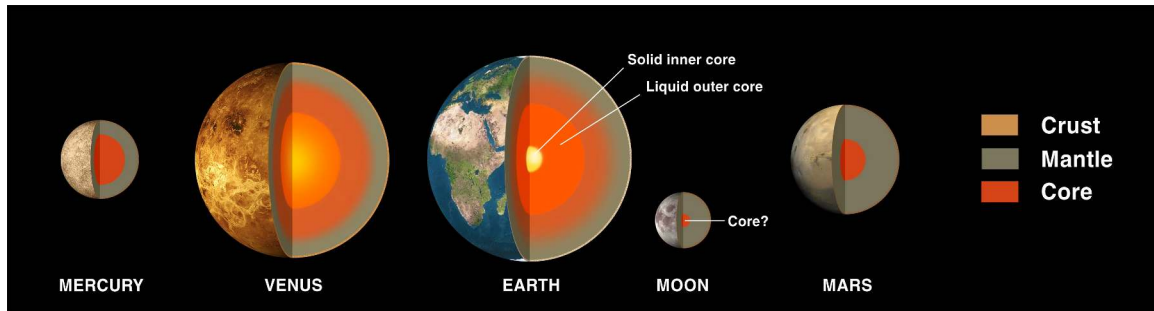
Mercury's layered interior from MESSENGER gravity and libration data. The metallic core extends to ~83% of the planet's radius and accounts for ~74% of its mass. After Margot et al. (2018).

Comparing terrestrial body interiors

Body	$\bar{\rho}$ (kg m ⁻³)	C/MR^2	R_c/R	Dynamo?
Moon	3346	0.393	~0.1–0.2	No (ancient)
Mars	3934	0.364	~0.49–0.54	No (ancient)
Venus	5243	~0.33	~0.55	No
Earth	5515	0.331	0.55	Yes
Mercury	5427	0.346	~0.83	Weak

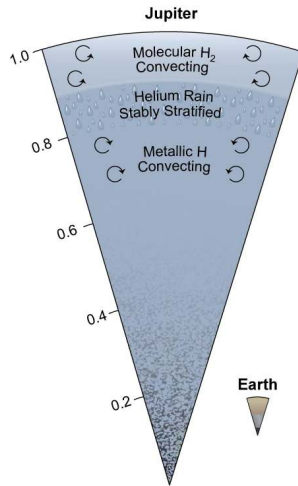
Mars: upper value from Stähler et al. (2021), lower from the 2023 reanalyses (Khan et al.; Samuel et al.); Mercury: Margot et al. (2018)

- Higher mean density $\Rightarrow$ larger iron core fraction
- Lower $C/MR^2 \Rightarrow$ stronger central mass concentration
- Dynamo requires convecting liquid conductor

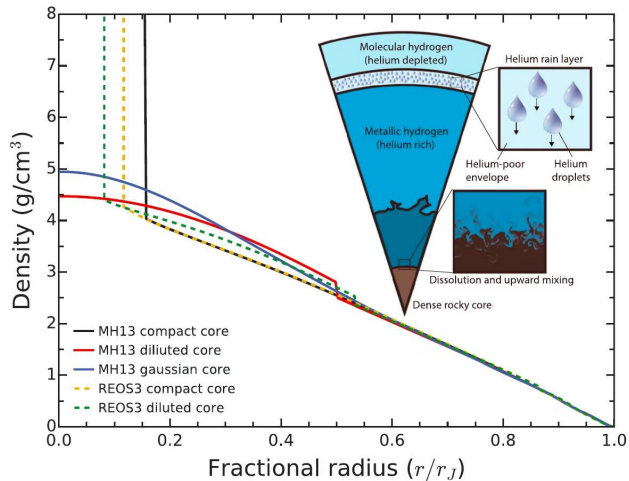


The terrestrial bodies in cross-section, at relative scale. All share a silicate mantle over a metallic core, but differentiation yields core fractions from ~10–20% to ~83%. Credit: NASA Solar System Exploration, public domain.

8. Giant planet interiors

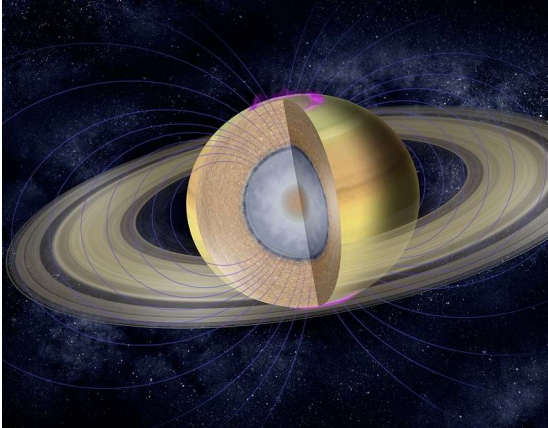


Jupiter's interior: molecular and metallic hydrogen above an extended, dilute core enriched in heavy elements. Credit: NASA/JPL-Caltech/SwRI/John E. Connerney (PIA25062).



Density profiles of representative Jupiter interior models from Juno gravity. Dilute-core solutions expand heavy elements out to half the planetary radius, contrasting with classical compact-core models. Credit: Wahl et al. (2017), Fig. 1.

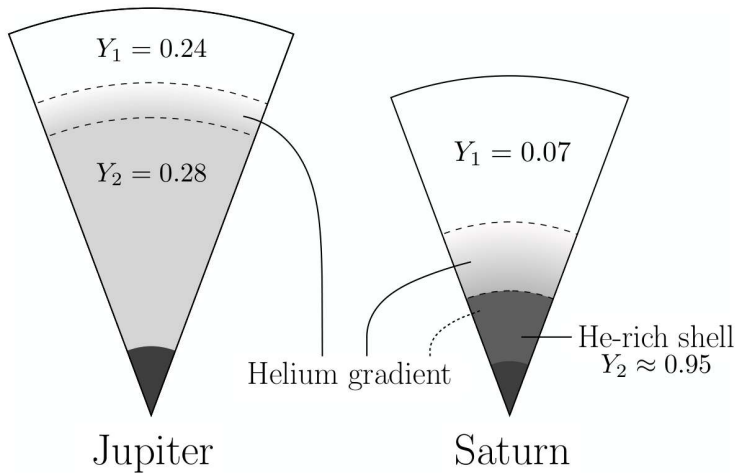
Saturn: helium rain



Credit: NASA/JPL-Caltech

Cooling inside Saturn induces helium immiscibility in metallic hydrogen, releasing gravitational energy that powers its excess infrared luminosity.

- Helium separates into droplets at $P \sim 2$ Mbar and rains toward the planet centre
- Sinking helium releases gravitational energy; with Kelvin-Helmholtz contraction it explains why Saturn radiates $\sim 1.8\times$ the solar flux it absorbs
- Ring seismology reveals a diffuse heavy-element core extending to $\sim 60\%$ of the radius



Helium phase separation in Jupiter and Saturn; sinking helium droplets release gravitational potential energy to explain Saturn's excess luminosity. Credit: Mankovich & Fortney (2020), Fig. 3.

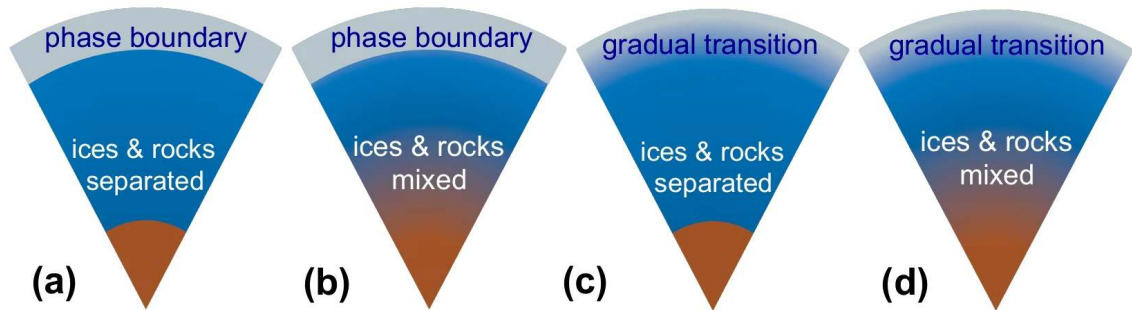


9. Ice giant interiors

Ice giants: canonical layers

Uranus and Neptune are modelled with a hydrogen-helium envelope, a thick conducting fluid ice mantle, and a rocky core.

- Outer H/He envelope holds ~20% of total mass
- Intermediate layer (~60–70% mass): hot fluid H_2O , NH_3 , and CH_4 at 10–500 GPa
- Deep water may reach an electrically conducting superionic state



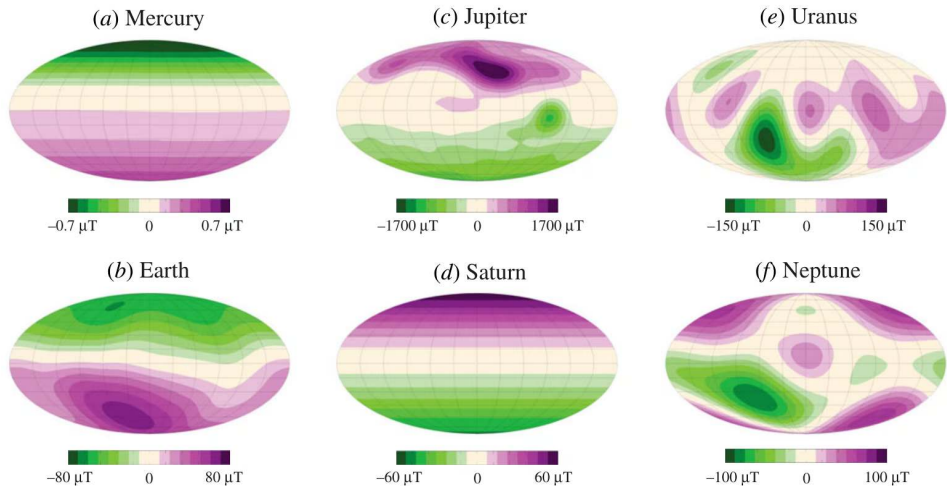
Four candidate internal structures of an ice giant. Gravity data alone cannot distinguish sharp layers from gradual compositional gradients, leaving their degree of differentiation open.

Credit: Helled et al. (2020), Fig. 4.

Ice giant magnetic fields: unusual geometry

Ice giants generate highly tilted, non-dipolar magnetic fields in thin, shallow conducting shells.

- Field axes are tilted (59° Uranus, 47° Neptune) and offset from centre
- Dynamos operate in thin outer fluid ice shells, not deep interiors
- Uranus emits negligible internal heat; Neptune emits $\sim 2.6\times$ solar flux



Radial magnetic field at the surface of six planets in Mollweide projection. Uranus and Neptune generate multipolar fields tilted by 59° and 47° and offset from the centre, unlike the axial dipoles of Earth, Jupiter, and Saturn. Credit: Soderlund et al. (2020), Fig. 1.

10. Icy moon interiors & ocean worlds



Ocean worlds: liquid water beneath ice

Tidal flexing and radiogenic decay sustain subsurface oceans beneath insulating ice shells:

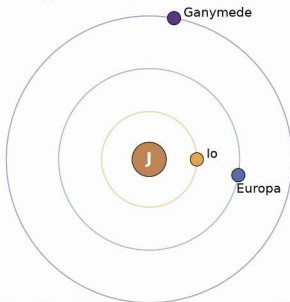
Moon	R (km)	ρ (kg m⁻³)	Ocean evidence
Europa	1561	3013	Magnetic induction, chaos terrain
Enceladus	252	1609	Plumes, libration
Titan	2575	1882	Tidal Love number k_2
Ganymede	2634	1936	Auroral oscillations



Europa's subsurface environment: an ice shell overlies a global ocean roughly 100 km deep in contact with a rocky seafloor. Credit: NASA/JPL-Caltech (PIA26438).

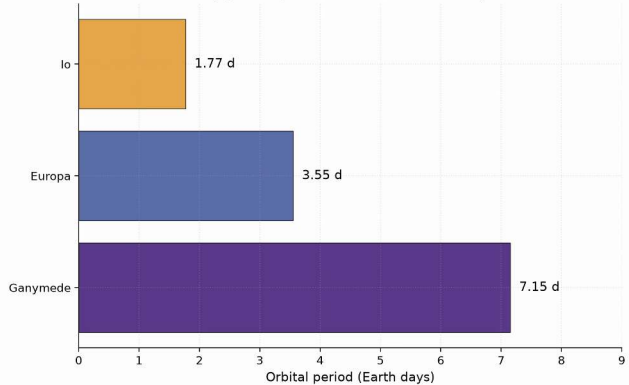
Laplace resonance (Io-Europa-Ganymede)

(a) Orbital configuration (not to scale)

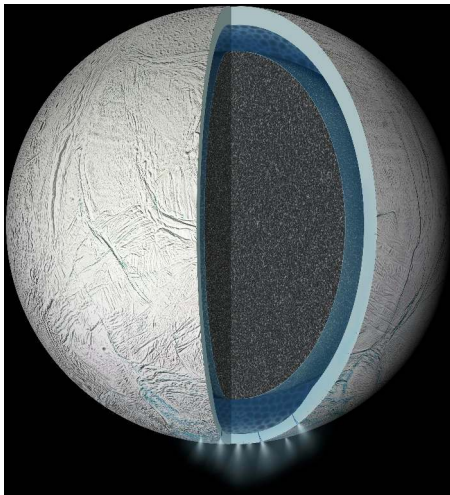


Mean-motion commensurability: $n_{\text{Io}} : n_{\text{Eur}} : n_{\text{Gan}} = 4 : 2 : 1$

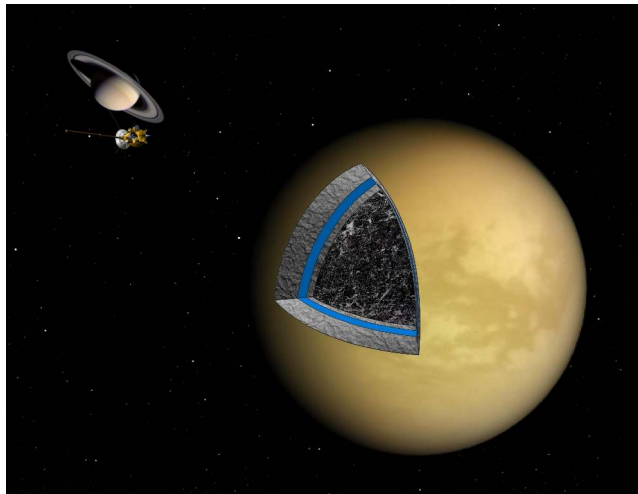
(b) Orbital periods: 1:2:4 commensurability



The Io-Europa-Ganymede 4:2:1 Laplace resonance forces non-zero eccentricities, driving tidal flexing that maintains Europa's liquid ocean; without the resonance, tides would circularise the orbit and the heating would stop. Course original; orbital periods from NASA/JPL Solar System Dynamics.



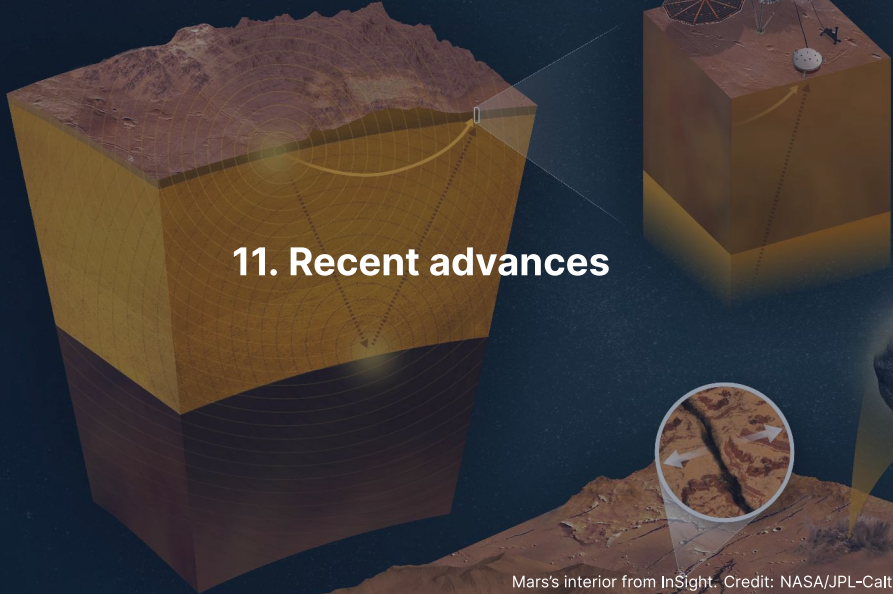
Enceladus in cutaway: a global ocean between an ice shell and porous core sustains active hydrothermal venting and south-polar plumes radiating ~16 GW. Credit: NASA/JPL-Caltech (PIA19656).



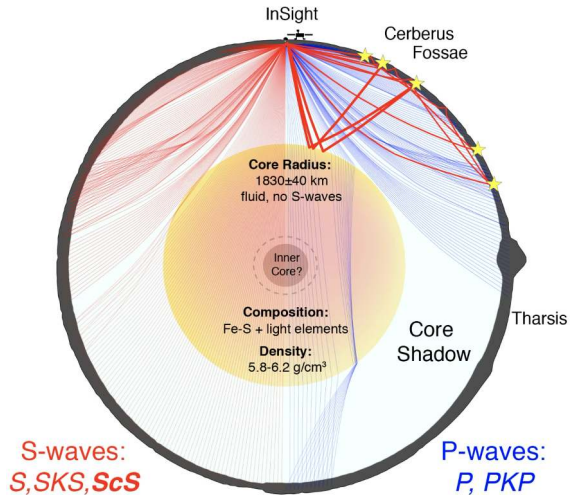
Titan's interior from Cassini gravity and radio science. A large tidal Love number k_2 indicates an interior ocean that decouples the outer ice shell from the high-pressure ice and rock below. Credit: NASA/JPL.

INSIDE MARS

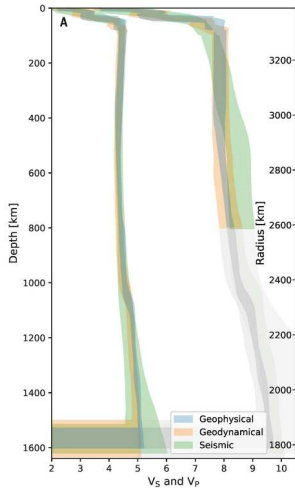
11. Recent advances



Mars's interior from InSight. Credit: NASA/JPL-Caltech (PIA25282)



Seismic ray paths from InSight across Mars delineate mantle propagation and the core shadow zone; the 2023 reanalyses put a molten silicate layer above a smaller core of ~1650–1675 km. Credit: Stähler et al. (2021), Fig. 3.



Mars velocity profiles from InSight show v_s dropping to zero at the core boundary, confirming a liquid metallic core. Credit: Stähler et al. (2021), Fig. 2A.

Summary

1. Seismology, gravity, and C/MR^2 probe deep interiors
2. $C/MR^2 = 0.400$ for uniform sphere; lower values indicate differentiation
3. Hydrostatic equilibrium and equations of state set radial profiles
4. Mantle phase transitions at 410 and 660 km govern convection
5. Tidal heating maintains subsurface oceans across icy worlds

Next: Lecture 9 turns to Earth and Venus: twins in size, contrasts in evolution.

Questions?

Next: Lecture 9, Rocky planets I: Earth & Venus

Mon 5 Oct, 15:00–17:00, room 5419.0161 (Collegezaal)

Next tutorial: Fri 2 Oct, 13:00–15:00, room 5419.0161 (Collegezaal)

<https://ips.formingworlds.space>

